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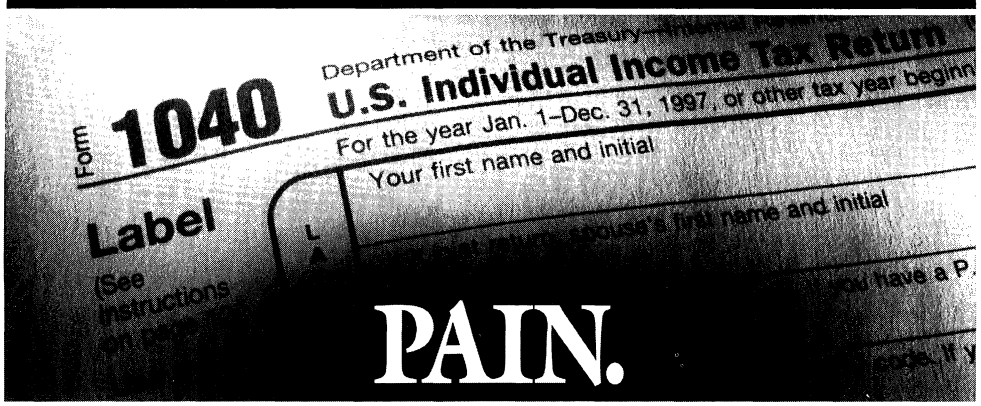

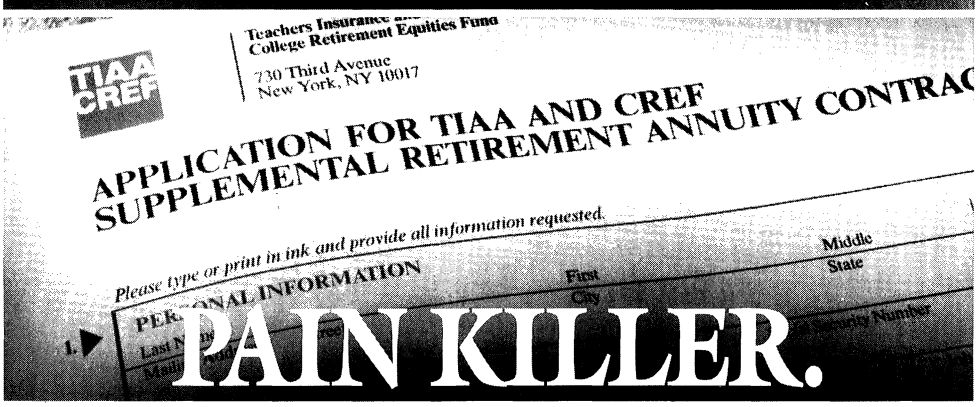

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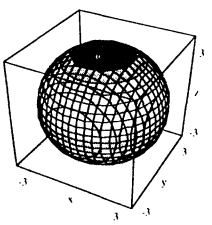

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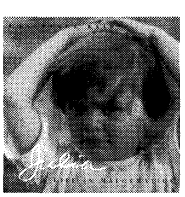

Julia a life in mathematics

Constance Reid

Constance Reid, an established writer about mathematicians, has written an excellent and loving book, about her sister Julia Robinson, the mathematician. The author has written that she wants the book to be one for all age groups and she has succeeded admirably in making it so...Julia wanted to be known as a mathematician, not a woman mathematician and rightly so! However, she was, and is, a wonderful role model for women aspiring to be mathematician. What a great gift this book would be!
—Alice Schafer, Former President, AWM

This book is a small treasure, one which I want to share with all my mathematical friends. The assembly of several articles and additional photos and remarks provides the image of a mathematician of extraordinary taste, tenacity and generosity.... Julia Robinson broke ground in displaying the deep connections between number theory and logic. Her results have led to a very active area today, making the appearance of this book very timely. Her work and her example are however timeless and I can think of no better advice to give a young mathematician, either in how to do mathematics. or how to behave in mathematics, than: "Be like Julia!"
—Carol Wood, Deputy Director, MSRI

Julia is the story of the life of Julia Bowman Robinson, the gifted and highly original mathematician who during her lifetime was recognized in ways that no other woman mathematician had been recognized up to that time. In 1976 she became the first woman mathematician elected to the National Academy of Sciences and in 1983 the first woman elected president of the American Mathematical Society.

This unusual book, profusely illustrated with previously unpublished personal and mathematical memorabilia, brings together in one volume the prizewinning "Autobiography of Julia Robinson" by her sister, the popular mathematical biographer Constance Reid, and three very personal articles about her work by outstanding mathematical colleagues.

All royalties from sales of this book will go to fund a Julia Robinson Prize in Mathematics at the high school from which she graduated.

Catalog Code: JULIA/JR
136 pp., Hardbound, 1996, ISBN 0-88385-520-8
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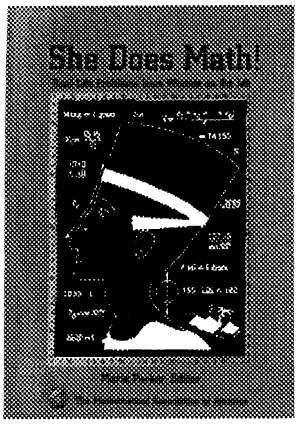

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She Does Math!

Real-Life Problems From Women On The Job

Marla Parker, Editor

She Does Math! presents the career histories of 38 professional women and math problems related to their work. Each history describes how much math the author took in high school and college; how she chose her field of study; and how she ended up in her current job. Each of the women presents problems that are typical of those she has faced in her job. The problems require only high school mathematics for their solution.

Who should have this book?

Your daughter, your granddaughter, your sister, your former math teacher, your students—and young men too. They want to know how the math they study is applied, and this book will show them.

There are many reasons to buy this book:

➤ By reading the career histories of the women profiled in this book, young people will learn that if they take mathematics courses in high school and college they will be qualified to enter interesting technical fields and earn good salaries.
➤ The problems have special appeal to students who are beginning to think about career choices.
➤ The book provides practical information about the job market in an interesting, innovative way.
➤ Strong female role models who work as successful technical professionals are presented.
➤ The problems are interesting and challenging, yet require only high school mathematics. They demonstrate how good math skills are applied to real-life problems.

Read what others have said about She Does Math!

Finally — a practical, innovative, well-written book that will also inspire its readers. The wonder is...it's a mathematics text and a biography! The idea of women telling their own career stories, emphasizing the mathematics they use in their jobs is extremely creative. This book makes me wish that I could go through school all over again!

Anne L. Bryant, Executive Director
American Association of University Women

She Does Math! *will undoubtedly appeal both to students who already enjoy math and want to get a view of potential career paths, and also to students who want to better understand the relevance of their math classes to their future careers. It is an absorbing look into the lives of some very inspiring and talented women!*

Susanne Hupfer and Elisabeth Freeman
Yale University

This collection is a wonderful confirmation that real women do math. They do math in a surprising variety of careers, fully enjoying the challenge and rewards of solving complex problems. This is a book for young women and men, a book for their teachers and parents, a book that informs about the possibilities that mathematics affords to all. It is also a book that will engage you in real-life mathematics!

– Doris Schattschneider
Moravian College

Catalog Code: SDM/JR98
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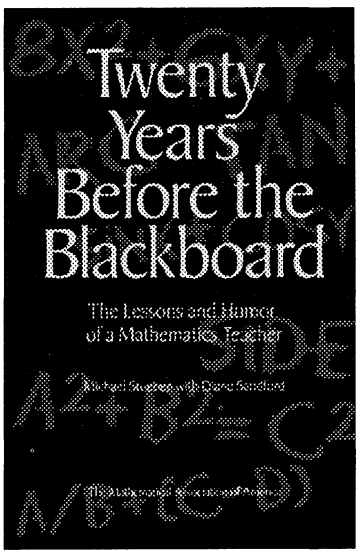

Twenty Years Before the Blackboard

The Lessons and Humor of a Mathematics Teacher

Michael Stueben with Diane Sandford

Series: Spectrum

A perfect gift for the new teacher. . . or for anyone interested in the teaching of mathematics.

This book is the legacy of twenty years of mathematics teaching. During this time, the author searched for motivation techniques, mnemonics, insightful proofs, and serious applications of humor to aid his teaching. The result is this book: part philosophy, part humor, and part biography. Readers will be amused and enlightened on every page.

Mr. Stueben shows how he has used humor and wordplay to motivate his students. The book is filled with wonderful problems and proofs, as well as the author's insights about how to approach teaching problem solving to high school students. Sections of the book also treat the use of calculators and computers in the classroom. A section on mnemonics shows how teachers can use memory aids to help their students learn and retain material.

All in all, *Twenty Years Before the Blackboard* provides a goldmine of ideas for the classroom teacher. Although Mr. Stueben taught at the high school level, his book is an excellent "methods" book for mathematics teachers at all levels.

Read what Martin Gardner has to say about this fascinating book:

It's been decades since I read so entertaining a book about mathematics. The book is a treasure-trove of mathematical jokes, rhymes, anecdotes, word play, mnemonics, and beautiful proofs. For teachers there is an abundance of wise advice based on the author's twenty years in high school teaching. Mathematicians at all levels, from amateurs to college professors will not only chuckle over its gems, but learn much they did not know before. —Martin Gardner

Catalog Code: TYB/JR98
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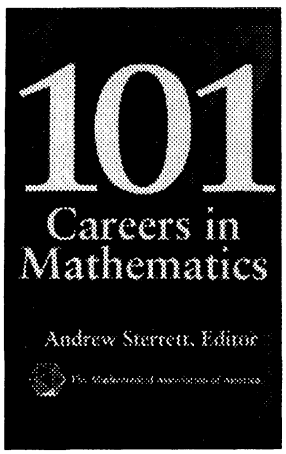

101 Careers in Mathematics

Andrew Sterrett, Editor

Series: Classroom Resource Materials

A career guide for your students. If they want to know why they should study mathematics, this book will tell them.

Read the biographical essays written by individuals who have gotten exciting good-paying jobs by preparing themselves with a solid background in the mathematical sciences. It will provide you and your students with a wealth of information about the types of different career paths that can be chosen for those who are well-prepared in mathematics.

These mathematicians are found:

- in well-known companies such as IBM, AT&T, and American Airlines,
- in some surprising places like FedEX Corporation, L. L. Bean, Perdue Farms,
- in government agencies
- in the arts (sculpture, music, and television),
- in the professions (law and medicine), and
- in education (elementary, secondary, college and university)

Many of these individuals have started their own companies.

Your students will see how these individuals use their mathematical sciences training on a daily basis in their work, often relying on the general problem-solving skills they have acquired in their mathematics courses. Those who studied statistics and computer science as well as mathematics, tell how their training in these disciplines helped them advance in their careers.

Articles in the Appendix reprinted from the MAA's magazine for students, *Math Horizons*, provide valuable advice on looking for a job and the expectations of industry.

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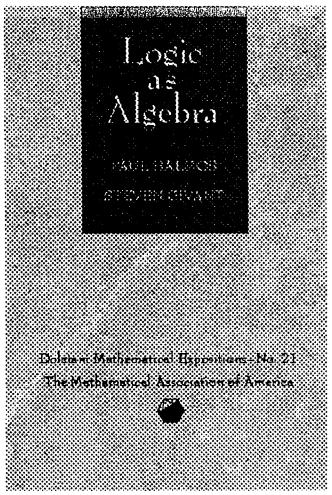

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Logic as Algebra

Paul Halmos and Steven Givant

Series: Dolciani Mathematical Expositions

This book is based on the notes of a course in logic given by Paul Halmos. This book retains the spirit and purpose of those notes, which was to show that logic can (and perhaps should) be viewed from an algebraic perspective. When so viewed, many of its principal notions are seen to be old friends, familiar algebraic notions that were "disguised" in logical clothing. Moreover, the connection between the principal theorems of the subject and well-known theorems in algebra becomes clearer. Even the proofs often gain in simplicity.

Propositional logic and monadic predicate calculus—predicate logic with a single quantifier— are the principal topics treated. The connections between logic and algebra are carefully explained. The key notions and the fundamental theorems are elucidated from both a logical and algebraic perspective. The final section gives a unique and illuminating algebraic treatment of the theory of syllogisms—perhaps the oldest branch of logic, and a subject that is neglected in most modern logic texts.

The presentation is aimed at a broad audience—mathematics amateurs, students, teachers, philosophers, linguists, computer scientists, engineers, and professional mathematicians. Whether the reader's goal is a quick glimpse of modern logic or a more serious study of the subject, the book's fresh approach will bring novel and illuminating insights to beginners and professionals alike. All that is required of the reader is an acquaintance with some of the basic notions encountered in a first course in modern algebra. In particular, no prior knowledge of logic is assumed. The book could serve equally well as a fireside companion and as a course text.

Contents: **What is Logic?:** To count or to think; A small alphabet; A small grammar; A small logic; What is truth?; Motivation of the small language; All mathematics. **Propositional Calculus:** Propositional symbols; Propositional abbreviations; Polish notation; Language as an algebra; Concatenation; Theorem schemata; Formal proofs; Entailment; Logical equivalence; Conjunction; Algebraic identities. **Boolean Algebra:** Equivalence classes; Interpretations; Consistency and Boolean algebra; Duality and commutativity; Properties of Boolean algebras; Subtraction; Examples of Boolean algebras. **Boolean Universal Algebra:** Subalgebras; Homomorphisms; Examples of homomorphisms; Free algebras; Kernels and ideals; Maximal ideals; Homomorphism theorem; Consequences; The representation theorem. **Logic via Algebra:** Pre-Boolean algebras; Substitution rule; Boolean logics; Algebra of the propositional calculus; Algebra of proof and consequence. **Lattices and Infinite Operations:** Lattices; Non-distributive lattices; Infinite operations. **Monadic Predicate Calculus:** Propositional functions; Finite functions; Functional monadic algebras; Functional quantifiers; Properties of quantifiers; Monadic algebras; Free monadic algebras; Modal logics; Monadic logics; Syllogisms.

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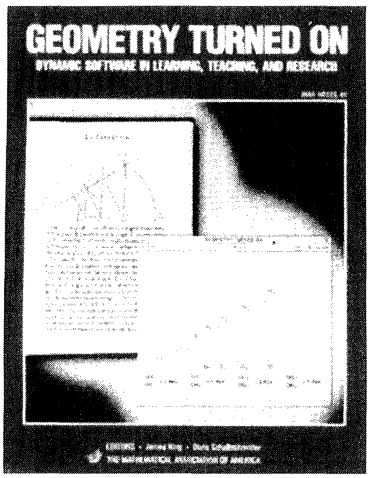

Geometry Turned On

Dynamic Software in Learning, Teaching and Research

JAMES KING AND DORIS SCHATTSCHNEIDER, EDITORS

Learn how you can use interactive computer software in the study and teaching of geometry

Series: MAA Notes

Dynamic geometry is active, exploratory geometry carried out with interactive computer software. It has had a profound effect on classroom teaching wherever it has been introduced and has become an indispensable research tool for mathematicians and scientists. The papers in this volume give a good idea of the ways in which the software can be used, and some of the effects it can have. It is clear that the software raises various questions for teaching and research, and its continuing evolution raises questions on the design of the software itself.

With the use of interactive computer software, the focus in teaching shifts from students laboriously making constructions by hand to verify a stated fact in a text (for which there seems little reason to produce a proof) to a focus on students carrying out experiments, quickly producing many accurate sketches from which they conjecture properties that seem to be "always" true.

The latest dynamic geometry software takes advantage of current computer hardware with mouse interface for graphics and great speed. What is most exciting about the software is its dynamic nature. Thus, after a geometric configuration is drawn, any unconstrained parts of the configuration (arbitrary segments or points, for example), that are not dependent on any other objects are moveable — they can literally be grabbed with a cursor (using the mouse) and can be dragged or stretched — and as they move, all other objects in the configuration automatically self-adjust, preserving all dependent relationships and constraints.

Although this volume is printed in a conventional manner, and every paper has illustrations, most of the illustrations beg to be played with. We want you to be able to experience some of the explorations described by our authors. To make that possible, dynamic sketches that use Geometer's Sketchpad or Cabri II have been made available by several of the authors and are posted on a Web page maintained by the Mathematics Forum.

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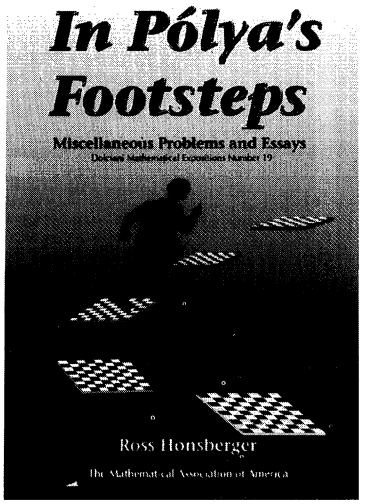

In Pólya's Footsteps

Miscellaneous Problems and Essays

Series: Dolciani Mathematical Expositions

Ross Honsberger

Another elegant collection of problems from Ross Honsberger

The study of mathematics is often undertaken with an air of such seriousness that it doesn't always seem to be much fun at the time. However, it is quite amazing how many surprising results and brilliant arguments one is in a position to enjoy with just a high school background. This is a book of miscellaneous delights, presented not in an attempt to instruct but as a harvest of rewards that are due good high school students and, of course, those more advanced — their teachers, and everyone in the university mathematics community. Admittedly, they take a little concentration, but the price is a bargain for such gems.

A half dozen essays are sprinkled among some hundred problems, most of which are the easier problems that have appeared on various national and international Olympiads. Many subjects are represented — combinatorics, geometry, number theory, algebra, probability, ... The sections may be read in any order. The book concludes with twenty-five exercises and their detailed solutions.

Something to delight will be found in every section — a surprising result, an intriguing approach, a stroke of ingenuity — and the leisurely pace and generous explanations make them a pleasure to read.

The inspiration for many of the problems came from the Olympiad Corner of *Crux Mathematicorum,* published by the Canadian Mathematical Society.

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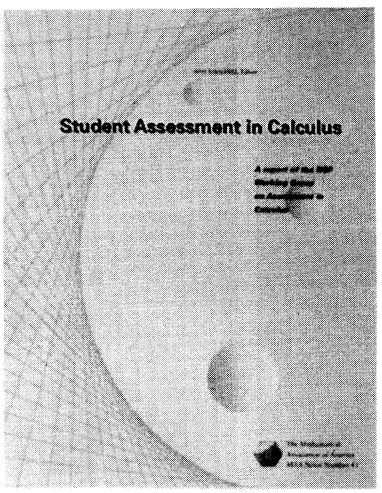

Student Assessment in Calculus

A Report of the NSF Working Group on Assessment in Calculus

ALAN SCHOENFELD, EDITOR
Series: MAA Notes

If you teach calculus, you should read this book. If you want to know what mathematics your students understand, or if you want to know how to find out what they understand, this book contains essential information for you.

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- develop a framework to tailor calculus instruction to the students' needs;
- establish an agenda for further research on student understanding;
- describe how to make use of a range of techniques to test what students know, such as multiple-choice tests or short essay questions, student portfolios and "clinical" interviews;
- summarize major goals of the reform movement and describe the challenges faced by those who are taking a closer look at how students learn;
- illustrate the ways in which calculus projects attempt (via exams, papers, projects, etc.) to find out what their students have learned.

This book is the result of those efforts. If you teach calculus, if you want to see examples of useful assessment techniques, or if you are interested in issues of how to measure student learning in mathematics, then there is a lot for you here.

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